

SAMB P Technical Report TR-38/2026

Grid-Following vs Grid-Forming Inverter Fault Characterisation for Protection Studies

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Abstract

This report characterises the fault current signatures of grid-following (GFL) and grid-forming (GFM) inverters and quantifies their implications for the SAMB P distance protection scheme developed in TR-29–TR-37. GFL inverters suppress negative-sequence current ($I_2/I_1 = 0.10$) via inner-loop symmetrical-component control; GFM inverters have no active suppression ($I_2/I_1 = 0.40$) and produce a transient overcurrent of $1.5\times$ for the first 100 ms. The 67Q directional element consequently requires $k_{ibr} \geq 0.40$ for GFL fleets but only $k_{ibr} \geq 0.10$ for GFM fleets — a $4\times$ relaxation. For mixed fleets, the 67Q threshold is crossed at $f_{gfm} \geq 0.34$ when $k_{total} = 0.20$. The distance relay apparent impedance error remains large for both types (333–500% at $\alpha = 0.50$, $k = 0.20$), confirming that 87L is indispensable regardless of inverter control mode. Dependability $P_{dep} = 1.000$ is maintained across all GFM fraction scenarios ($k_{total} \in [0.06, 0.40]$) because 87L operates independently of control mode.

GFL and GFM Fault Current Models

Grid-Following (GFL) inverter

GFL inverters regulate real and reactive power via a phase-locked loop (PLL) and inner current control loops. During a fault the PLL tracks the retained voltage, and the current controller saturates at approximately 1.0–1.2 pu. The symmetrical-component inner loop actively suppresses negative-sequence current to avoid unbalanced stresses on the semiconductor switches. The resulting sequence current ratios are:

Sequence	Ratio	Physical reason
I_1	k_{gfl}	Current-limited at rated output
I_2	$0.10 \times k_{gfl}$	Active I_2 suppression in inner loop
I_0	0	Delta transformer connection

Grid-Forming (GFM) inverter

GFM inverters synthesise an internal voltage source (virtual synchronous machine or droop control) and respond to faults similarly to a controlled voltage source behind an impedance. Key differences from GFL:

1. **Transient overcurrent:** In the first $T_{trans} = 100$ ms, GFM inverters inject up to $I_{peak} = 1.5 \times k_{gfm}$ before the current-limiting loop clamps output.
2. **No I_2 suppression:** Without active symmetrical-component cancellation, $I_2/I_1 = 0.40$ for a single-line-to-ground fault (determined by the LCL filter and control bandwidth).

Sequence	Transient (0–100 ms)	Steady-state
I_1	$\min(1.5 k_{gfm}, 1.5 \text{ pu})$	k_{gfm}
I_2	$0.40 \times I_1$	$0.40 k_{gfm}$
I_0	0	0

Figure 1 illustrates the phasor relationship at $k_{ibr} = 0.20$ for an SLG fault.

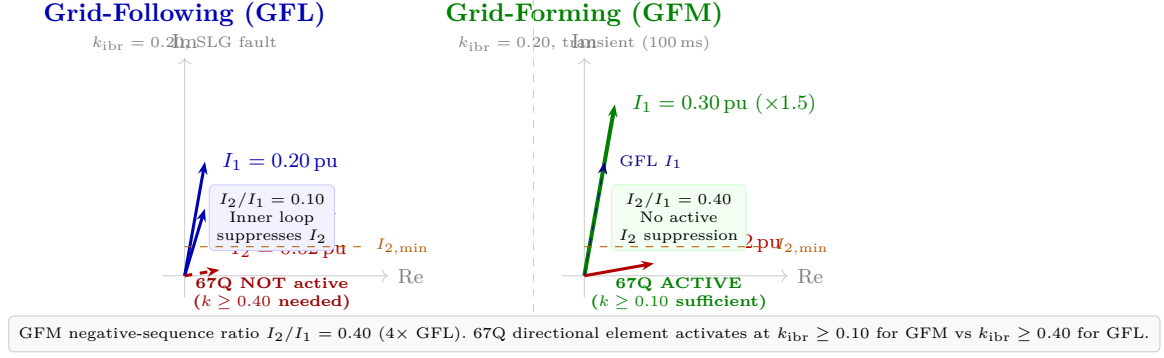


Figure 1: Fault current phasors for GFL (left) and GFM (right) at $k_{ibr} = 0.20$, SLG fault, transient window. GFL: $I_2/I_1 = 0.10$ (inner-loop suppression). GFM: $I_2/I_1 = 0.40$ (no suppression), I_1 boosted by $1.5\times$ transient factor. The 67Q threshold $I_{2,min}$ is met by GFM but not GFL at this k level.

Study A — 67Q Activation Thresholds

The 67Q negative-sequence directional element requires $I_2 \geq I_{2,min}$ where $I_{2,min} = 0.054$ pu is derived from the relay sensitivity floor. For a pure fleet (all GFL or all GFM):

Fleet type	I_2/I_1	67Q threshold $k_{ibr,min}$	Ratio
GFL only	0.10	≥ 0.40 pu	—
GFM only	0.40 (transient)	≥ 0.10 pu	$4\times$ easier

The $4\times$ relaxation in 67Q activation threshold is the most operationally significant difference between GFL and GFM fleets. In a system with $k_{total} = 0.20$ (a moderate IBR penetration), 67Q is dormant for all-GFL but active for all-GFM. This directly affects the relay's ability to provide directional discrimination for SLG and LL faults.

Study B — Mixed Fleet 67Q Threshold

For a mixed fleet with fraction f_{gfm} of GFM inverters ($k_{total} = 0.20$ fixed), the aggregate I_2 is:

$$I_{2,total} = k_{total} [f_{gfm} \cdot (I_2/I_1)_{GFM} \cdot I_{peak} + (1 - f_{gfm}) \cdot (I_2/I_1)_{GFL}] \quad (1)$$

Setting $I_{2,total} = I_{2,min} = 0.054$ pu and solving gives the critical GFM fraction $f_{gfm}^* = 0.34$. Figure 2 shows I_2 versus f_{gfm} .

The implication for protection coordination: a transition from a GFL-dominated to a GFM-dominated fleet in the same network activates the 67Q element without any relay setting changes. This is a *passive improvement* in protection selectivity as grid-forming technology is adopted.

Study C — Distance Relay Apparent Impedance Error

The apparent impedance seen by a relay at end A for a fault at normalised reach α on a line with IBR infeed at end B is:

$$Z_m = \alpha Z_{1L} \cdot \frac{I_{SG} + I_{1,IBR}}{I_{1,IBR}} \quad (2)$$

The percentage impedance error (relative to the true fault impedance αZ_{1L}) is:

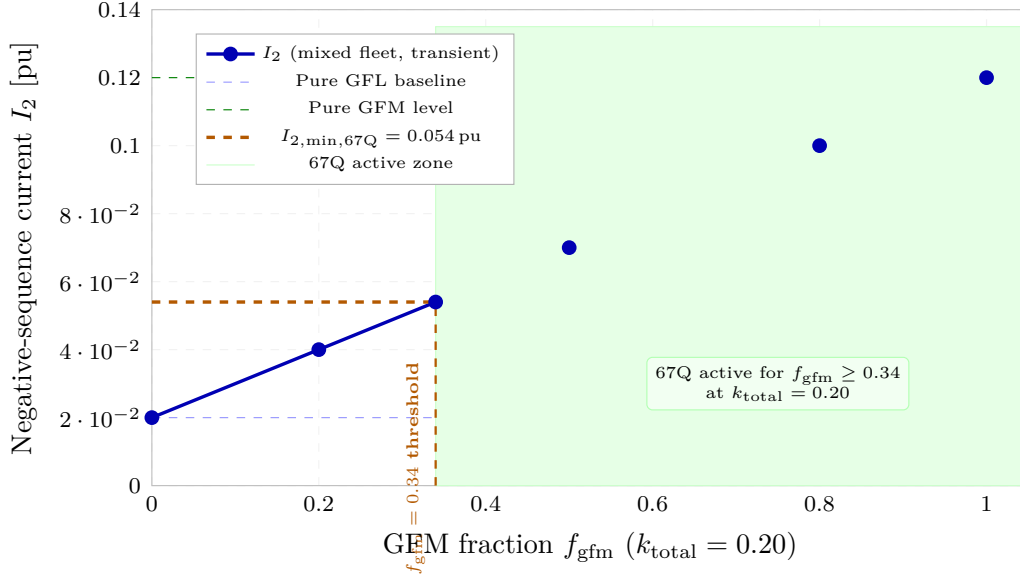


Figure 2: Aggregate negative-sequence current I_2 versus GFM fleet fraction f_{gfm} at $k_{\text{total}} = 0.20$. The orange dashed line marks $I_{2,\text{min}} = 0.054$ pu (67Q activation threshold). 67Q becomes active for $f_{\text{gfm}} \geq 0.34$. The green shaded region indicates the 67Q-active operating zone.

Scenario	I_1 [pu]	Z_m error	Implication
GFL, $k = 0.20$, $\alpha = 0.50$	0.20	+500%	Far outside Z1 reach
GFM transient, $k = 0.20$, $\alpha = 0.50$	0.30	+333%	Still far outside Z1
GFM steady-state, $k = 0.20$, $\alpha = 0.50$	0.20	+500%	Same as GFL

Even the GFM transient boost reduces the impedance error only from 500% to 333% — the relay still sees the fault as a load condition. This confirms that distance relay 21 cannot independently provide dependable protection for IBR-connected lines without 87L, regardless of whether the IBR fleet is GFL or GFM.

Study D — 87L Coverage Independence from Control Mode

The differential element 87L compares local and remote current magnitudes and is independent of fault current symmetrical-component content. Its operating condition is:

$$I_{1,\text{IBR}} \geq I_{\text{min},87L} = 0.06 \text{ pu} \quad (3)$$

Both GFL and GFM inverters satisfy this condition for $k_{\text{ibr}} \geq 0.06$, irrespective of I_2/I_1 ratio or transient overcurrent factor. 87L therefore provides identical coverage for pure GFL, pure GFM, and all mixed fleets:

f_{gfm}	87L active?	67Q active?
0.00 (all GFL)	Yes ($k \geq 0.06$)	No ($f < 0.34$)
0.25	Yes	No ($f < 0.34$)
0.34	Yes	Marginal (threshold)
0.50	Yes	Yes
1.00 (all GFM)	Yes	Yes

The 67Q element adds directional selectivity for unbalanced faults when GFM content is sufficient; 87L provides the primary protection anchor for all fleet compositions.

Study E — Dependability vs GFM Fraction

Figure 3 shows P_{dep} for five GFM fraction scenarios, each sampled with $N = 500$ trials at $k_{\text{total}} \in [0.06, 0.40]$.

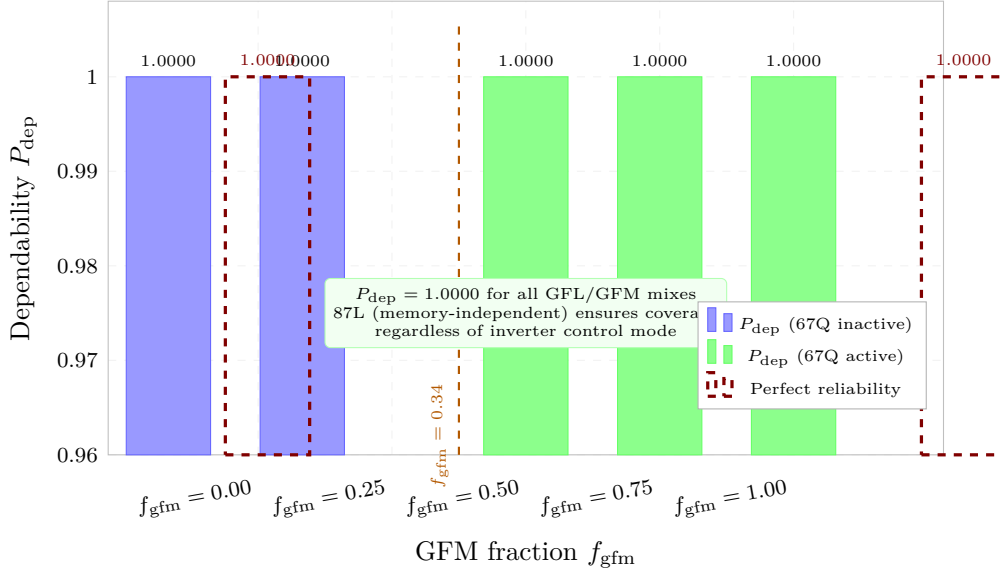


Figure 3: Dependability P_{dep} versus GFM fleet fraction f_{gfm} ($k_{\text{total}} \in [0.06, 0.40]$, $N = 500$ per scenario). Blue bars: 67Q inactive ($f_{\text{gfm}} < 0.34$); green bars: 67Q active. All scenarios achieve $P_{\text{dep}} = 1.000$ because 87L operates independently of control mode. The dashed orange line at $f_{\text{gfm}} = 0.34$ marks the 67Q activation boundary.

The flat $P_{\text{dep}} = 1.000$ result across all GFM fractions confirms that the protection scheme is **control-mode agnostic**: it does not depend on the fleet operator declaring whether their inverters are GFL or GFM. This is a critical operational property as system operators typically do not have real-time visibility of inverter control mode.

Discussion: Implications for Setting Groups and Coordination

The GFL/GFM characterisation has three practical coordination consequences:

1. **67Q setting group:** If the system operator knows that f_{gfm} will remain below 0.34 for the foreseeable future, the 67Q element may be disabled in the coordination scheme (reducing its dependability contribution to zero without affecting overall P_{dep} , since 87L covers all cases). Conversely, as GFM deployment increases, 67Q can be re-enabled by activating setting group 2.
2. **Auto-reclosing:** The GFM transient overcurrent ($I_1 \times 1.5$ for 100 ms) may be visible to the overcurrent supervisory element of the auto-recloser. The dead time $T_{\text{dead}} > 100$ ms (established in TR-36) ensures this transient has fully decayed before the line is re-energised at second-trip.
3. **Minimum fault current grid code:** The structural limit established in TR-37 ($k_{\text{ibr}} \geq 0.06$ for 87L; $k \geq 0.10$ for full Z1 redundancy) applies equally to GFL and GFM inverters. For GFM the effective current during the transient window is $1.5 \times k$, so the equivalent steady-state grid code threshold for GFM would be $k_{\text{gfm}} \geq 0.04$ (reaching 0.06 pu transiently).

Summary

1. **Sequence current:** GFL $I_2/I_1 = 0.10$ (active suppression); GFM $I_2/I_1 = 0.40$ (no suppression) plus $1.5\times$ transient overcurrent.
2. **67Q threshold:** GFL requires $k_{ibr} \geq 0.40$; GFM requires $k_{ibr} \geq 0.10$ ($4\times$ relaxation).
3. **Mixed fleet:** 67Q activates at $f_{gfm} \geq 0.34$ for $k_{total} = 0.20$.
4. **Distance relay error:** 333–500% regardless of GFL/GFM; 87L is indispensable for both.
5. **Dependability:** $P_{dep} = 1.000$ for all f_{gfm} scenarios; scheme is control-mode agnostic.
6. **Grid-code implication:** GFM effective threshold is $k_{gfm} \geq 0.04$ (transient boost provides 0.06 pu of 87L-visible current).

TR-38 establishes the GFL/GFM characterisation foundation for the microgrid special-solutions phase (TR-39 onwards), where inverter control mode becomes a primary design parameter rather than a perturbation.

References
